

Young Innovators · Class 9

Rechargeable Power Pack for Robots

Charge it, measure it, regulate it

Lithium cells

Safe charging

Voltage divider

Buck converter

Did you know?

The lithium-ion battery won the 2019 Nobel Prize in Chemistry. It powers phones, laptops and electric cars.



Li-ion cell (protected)



TP4056 charger





Let's think...

Why do robots keep dying mid-race?



Flat again!

Our robots eat AA batteries. They get weaker every minute, the motors slow down, and then we throw the batteries away. Phones charge hundreds of times. Could our robots work like a phone?



Li-ion cell (protected)



Bigger batteries?

Plug into the wall?

Rechargeable pack!

Today: you build a **rechargeable power pack** with a charger, a battery gauge on an OLED, and a regulated output.





Learning objectives

By the end of this class I can...



1



Battery maths: explain **voltage**, **capacity (mAh)** and **energy (Wh)** of a lithium cell

Proof: I can calculate how long my robot runs on one charge

2



Charge safely: charge an 18650 cell safely with a **TP4056** module

Proof: the charger LED turns from red to blue/green and I followed every safety rule

3



Measure: measure battery voltage with a **voltage divider** and show the % on an OLED

Proof: my OLED reading is within 0.1 V of the multimeter

4



Regulate: set a **buck converter** to a steady output voltage

Proof: the multimeter shows $3.3\text{ V} \pm 0.05\text{ V}$ at the output





Objective 1 · Battery maths · video

Video time!



How a Lithium-Ion Battery Works

Owl WiS · about 3 minutes · youtu.be/YFd0kb9Nwt0

Watch for...

- 1 Which way do the **lithium ions** move when the battery is charging?
- 2 What are the **anode** and **cathode** made of?
- 3 What does the **separator** do, and why is a damaged one dangerous?



After the video: Explain to your partner, using your hands as the two electrodes, what happens when you charge and then use a battery.





Objective 1 · Battery maths

Volts, mAh and watt-hours



The idea

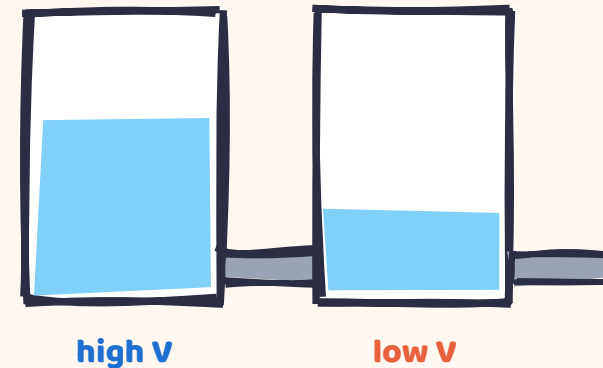
Voltage (V): how hard the battery pushes. One lithium cell: **4.2 V** full, about **3.7 V** in normal use, **3.0 V** empty.

Capacity (mAh): how much charge it holds. 2500 mAh = 250 mA for 10 hours.

Energy (Wh) = V × Ah: $3.7 \times 2.5 = 9.25 \text{ Wh}$.

Run time ≈ capacity ÷ current. Robot draws 500 mA → $2500 \div 500 = 5 \text{ hours}$ (less in real life).

Example: a water tank



Voltage is like **how high** the tank is (more pressure). Capacity is **how much water** it holds. A tall thin tank and a short wide tank can hold the same water but push it out differently.





Real world

Battery packs everywhere



Phones

One lithium cell, about 4000 to 5000 mAh



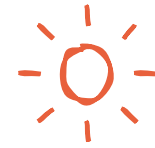
Electric cars

Thousands of 18650-style cells joined together



Drones

Light lithium-polymer packs give 20 to 40 min of flight



Solar homes

Store daytime solar energy for the night



In Pakistan, many homes use UPS batteries during load-shedding. What kind of battery is inside, and how long does it last?

Bigger is not always better
More capacity means more weight. Drone engineers fight for every gram.



Class game

Battery budget



How to play

1. Each team gets a pretend **2000 mAh** battery (20 paper tokens of 100 mAh).
2. Teacher reads out robot jobs: **motors 1 hour = 600 mAh, LEDs 1 hour = 40 mAh, Wi-Fi 1 hour = 150 mAh, OLED 1 hour = 20 mAh...**
3. Plan a robot's day: pay tokens for each job.
4. Which team can run its robot longest and still do something useful?

Then discuss

What used the most energy? Which jobs could you switch off or slow down?

Big idea

Engineers design for energy. Motors and radios are usually the hungriest parts.



Li-ion cell (protected)

2000 mAh



Safety first

Lithium cells store a LOT of energy



Never short-circuit + to – with a wire or metal can make it very hot in seconds.



Never damage don't drop, crush, pierce, or bend a cell. Swollen or dented? Give it to the teacher.



Use the right charger only the TP4056 module, never a direct USB wire. Check polarity before connecting.



Watch it never charge unattended or under books. Touch test: warm is OK, hot means unplug.



Recycle old cells go to battery recycling, not the bin.



Li-ion cell (protected)



Teacher check

Every circuit is checked by the teacher BEFORE the cell goes in the holder.





Build it

Our power kit (per group)



18650 cell + holder

The rechargeable energy store



TP4056 module

Safe lithium charger



LM2596 buck

Steps voltage down



Arduino + OLED

Battery gauge display



Multimeter

Checks every voltage



2 × 10k resistors

Voltage divider

Also: breadboard, jumper wires, LED + 220 Ω and buzzer (low battery alarm).

Protected module

Use the TP4056 version with 6 pins (B+, B-, OUT+, OUT-). It also cuts off if the cell gets too empty.



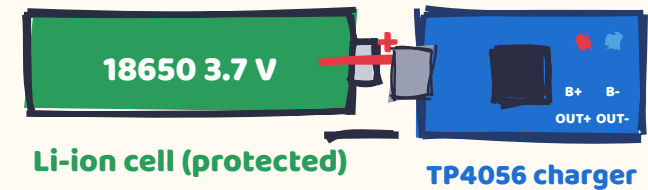


Objective 2 · Build step 1 of 3

Charge the cell



- With the **cell OUT**, check the holder's red wire is +. Multimeter on DC volts.
- Holder + → **B+**, holder – → **B–** on the TP4056.
- Teacher check. Then put the cell in, **flat end to –**.
- Plug a USB cable into the TP4056. Red LED = charging, blue/green = full.
- Measure the cell every 10 minutes and record it in the table on the next test slide.



Check: Voltage rises towards 4.2 V and the LED turns blue/green when full.



Gets hot, smells or hisses? Unplug USB, step back and call the teacher.





Objective 3 · Measure

A voltage divider measures safely



The idea

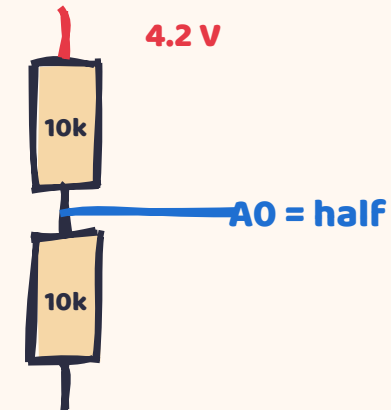
Two equal resistors in series split a voltage in **half**. Battery up to 4.2 V → the Arduino sees up to **2.1 V** on A0.

In code: `volts = analogRead(A0) × 5.0 ÷ 1023 × 2`
(× 5 ÷ 1023 turns the reading into volts, × 2 undoes the halving.)

Common ground

Battery – must also connect to Arduino GND, or the reading is nonsense.

Example: sharing a pizza fairly



Two equal friends share a pizza: each gets **half**. Two equal resistors share the voltage the same way. If one friend were twice as hungry (twice the resistance), they would get two thirds.





Objective 3 · Measure · Build step 2 of 3

Battery gauge on the OLED



```
#include <Wire.h>
#include <Adafruit_SSD1306.h>
Adafruit_SSD1306 oled(128, 64, &Wire, -1);

void setup() {
  oled.begin(SSD1306_SWITCHCAPVCC, 0x3C);
  oled.setTextColor(SSD1306_WHITE);
}

void loop() {
  float v = analogRead(A0) * 5.0 / 1023.0 * 2; // undo divider
  int pct = constrain((v - 3.0) / 1.2 * 100, 0, 100);

  oled.clearDisplay();
  oled.setTextSize(2);
  oled.setCursor(0, 0); oled.print(v, 2); oled.print(" V");
  oled.setCursor(0, 22); oled.print(pct); oled.print(" %");
  oled.drawRect(0, 46, 102, 14, SSD1306_WHITE);
  oled.fillRect(1, 47, pct, 12, SSD1306_WHITE);
  oled.display();

  if (v < 3.3) tone(8, 1000, 100); // low battery beep
  delay(1000);
}
```

Wiring

OLED SDA → A4, SCL → A5, VCC → 5V.
Divider middle → A0. Buzzer on pin 8.

Rough %

Lithium voltage does not fall in a straight line, so this % is an **estimate**. Phones use smarter 'fuel gauge' chips.



Check

Compare with the multimeter. Off by more than 0.1 V? Measure your real 5 V pin and use that instead of 5.0.





Objective 4 · Regulate

A buck converter steps voltage down



The idea

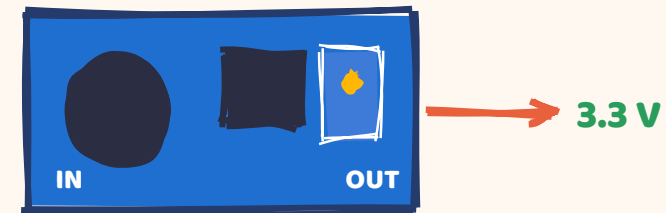
Different parts want different voltages: ESP32 **3.3 V**, Arduino and servos **5 V**, motors **6 to 12 V**.

A **buck converter** switches on and off very fast to turn a higher voltage into a lower, steady one, wasting little energy (about 80 to 90% efficient). Turn the small screw to set the output.

Only down!

A buck only goes DOWN. One 18650 (3.0 to 4.2 V) cannot make 5 V with it: you would need a BOOST converter or two cells.

Example: a dam with gates



LM2596 buck

A dam holds a high lake, but the gates let out a steady, gentle river. Open them more or less to control the flow. The buck converter is the gate between a high battery voltage and your delicate chip.



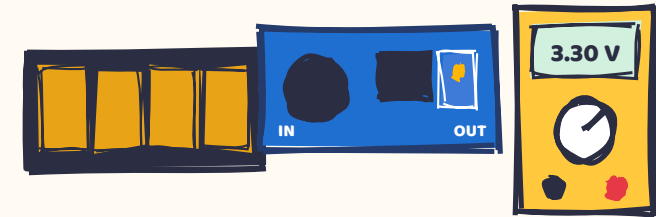


Objective 4 · Build step 3 of 3

Set a steady output



- a** Connect the 4×AA pack (about 6 V) to the buck **IN+** and **IN-**. Nothing on the output yet!
- b** Multimeter on the **OUT** pins. Turn the screw slowly.
- c** Set exactly **3.30 V**. Clockwise usually lowers it on most LM2596 boards.
- d** Add the LED + 220 Ω as a load. Does the voltage stay steady?
- e** Measure input and output. Which voltage changed when the batteries were swapped?



Check: Output stays at 3.3 V even when the input changes from 6.2 V to 5.5 V.



Always set the voltage **BEFORE** connecting a board. 6 V into an ESP32 destroys it.





Test like an engineer

Record your results



Measurement	Value	Unit
Cell at start of charging		V
After 10 / 20 / 30 min		V
Full (LED changed)		V
OLED reading vs multimeter		difference
Buck output, no load / with LED		V

Debug checklist

OLED shows 0 V? Check the divider middle goes to A0 and battery – to GND.

OLED blank? Try address 0x3D.



Charging slows down near 4.2 V. That is normal: the charger switches from constant current to constant voltage.





Level up!

Power challenges



Choose your challenge



Run time: measure your robot's current with the multimeter and predict its run time.



Boost: research how an MT3608 boost converter could give 5 V from one cell.



Solar charge: feed the TP4056 from a 6 V solar panel. How long to fill?



Better %: use a lookup table (4.2=100, 3.9=75, 3.8=50, 3.7=25, 3.4=5) instead of a straight line.

Think like an engineer

Why do phones say 1% for a long time, then die suddenly?



Boss level

Build a complete pack: cell + TP4056 + boost + gauge in one box, and power last week's robot.





Objective check

Can I really do it? Prove it.



01 A robot uses 400 mA from a 2000 mAh cell. About how long will it run?

☐☐☐

02 Name three safety rules for lithium cells.

☐☐☐

03 Show your OLED gauge and the multimeter. How close are they?

☐☐☐

04 Show the buck output on the multimeter. Why can't it make 5 V from one 18650?

☐☐☐

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Power audit: every team states its robot's run time, then tests it next week.





Show what you know

What did we learn?

- ☒ Lithium cells: 4.2 V full, 3.7 V normal, 3.0 V empty
- ☒ Run time $\approx$ capacity (mAh) $\div$ current (mA)
- ☒ Charge only with a proper module, never unattended
- ☒ A voltage divider lets the Arduino measure safely
- ☒ A buck converter steps voltage down, steadily



Exit ticket: your gauge shows 3.2 V. Is the cell nearly full or nearly empty, and what should happen?

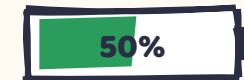


At home

Find the mAh rating on a phone or power bank at home. How many times could it charge your robot?



Li-ion cell (protected)



Next class:
Wireless
Doorbell with
Visitor Counter!

